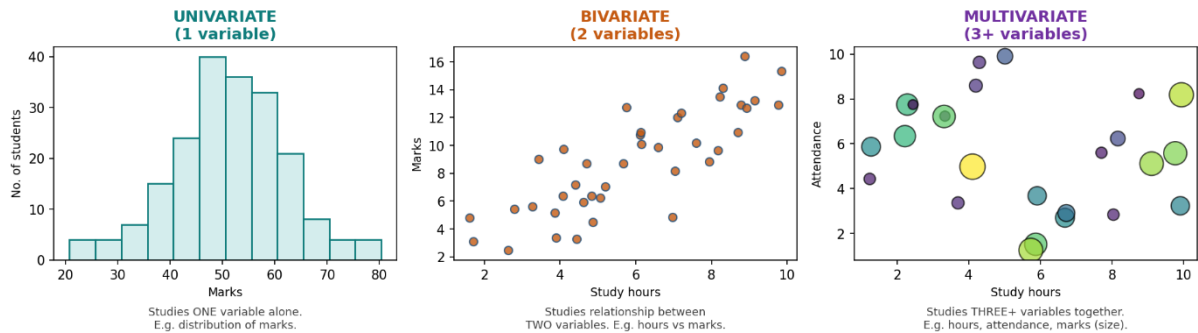


1. Explain data exploration in detail with example. ★ ★

Univariate vs Bivariate vs Multivariate Analysis



Data Exploration means **looking at and understanding data before doing any analysis**. It is the first step where we get familiar with the data — its size, type, patterns, and problems.

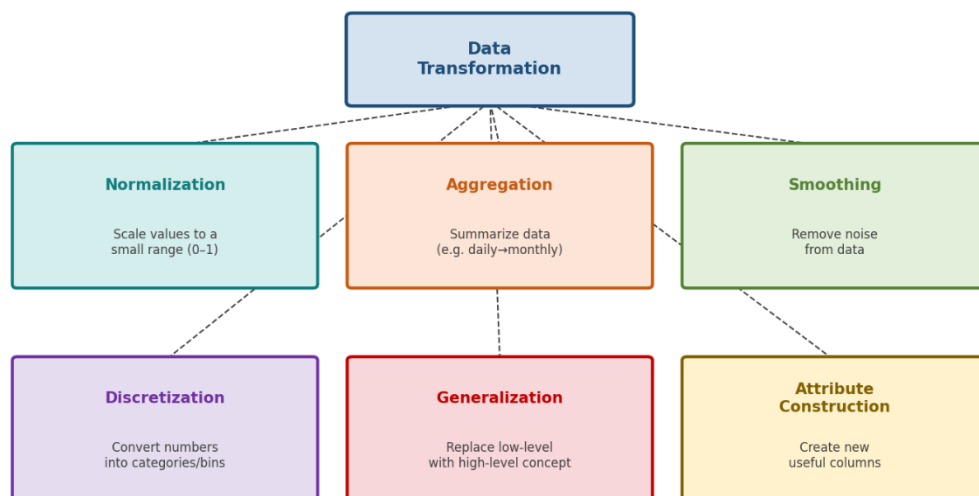
What we do in data exploration:

- Check the **structure** — how many rows and columns, and what each column means.
- Find **summary values** — like average, minimum, maximum, and count.
- Use **charts** (bar, histogram, scatter) to see patterns and trends visually.
- Spot **problems** — missing values, odd values (outliers), or errors.

Example: Before analyzing student marks, we explore the data: total 200 students, average mark 50, highest 95, lowest 20, and a histogram shows most students scored between 40 and 60.

Why useful: It helps us understand the data, find data-quality issues early, and decide how to clean and analyze it.

2. Explain data transformation in detail with example. ★ ★ ★ ★



Data Transformation means **changing data into a suitable form** so it can be analyzed easily and correctly. Raw data is often in different formats and scales, so we convert it into a clean, common format.

Main techniques (with simple examples):

1. **Normalization** — Scaling values into a small fixed range (like 0 to 1) so that big numbers don't dominate small ones.
 - *Example:* Salary (₹20,000–₹2,00,000) and Age (20–60) are scaled to 0–1 so both are treated fairly.
2. **Aggregation** — Combining detailed data into summary values.
 - *Example:* Daily sales added up to get **monthly** sales.
3. **Smoothing** — Removing noise (random errors) from data to make it cleaner.
 - *Example:* Replacing a wrong sudden spike with a smooth value.
4. **Discretization** — Converting continuous numbers into categories (bins).
 - *Example:* Age 27 → "Young Adult" group.
5. **Generalization** — Replacing low-level data with a higher-level concept.
 - *Example:* "Pune, Mumbai" → "Maharashtra".
6. **Attribute Construction** — Creating new useful columns from existing ones.
 - *Example:* From "Height" and "Weight", create a new column "BMI".

Why important: It makes data consistent, removes errors, and prepares it so analysis gives correct and meaningful results.

3. Explain data validation, incompleteness, noise, and inconsistency of quality of input data. ★ ★ ★

When data comes in, its **quality must be checked**. Poor-quality data leads to wrong results. The main quality problems are:

1. Data Validation Checking whether data is **correct, complete, and in the right format** before using it.

- *Example:* Checking that an "Age" column has only positive numbers, and an email has the "@" symbol.

2. Incompleteness (Missing Data) Some values are **missing or blank**.

- *Example:* A customer record has no phone number.
- *Fix:* Fill with average value, fill manually, or remove the record.

3. Noise Random errors or meaningless values in the data, often wrong or extreme.

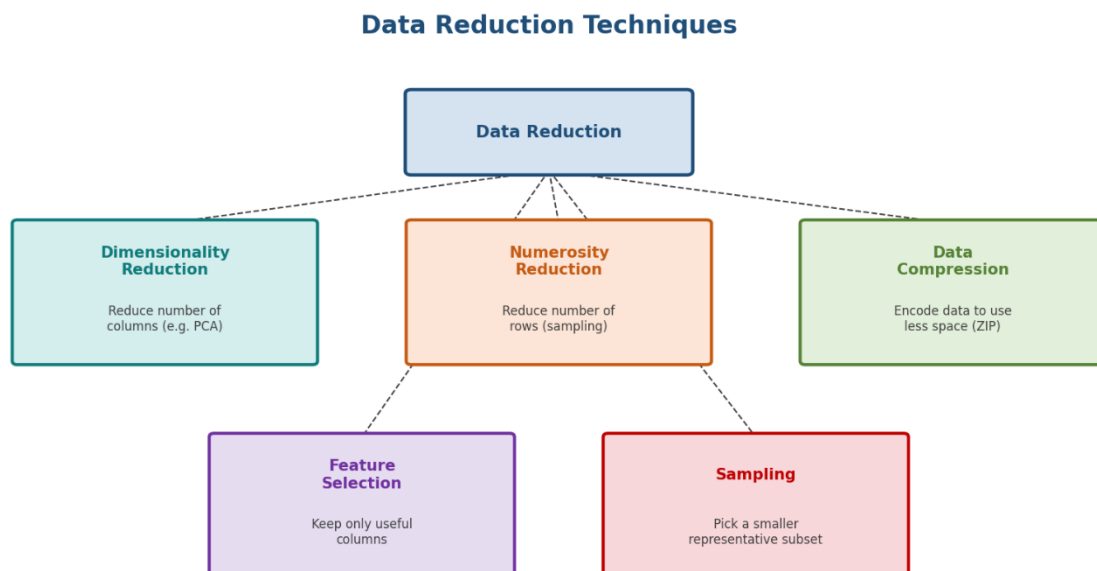
- *Example:* A person's age recorded as 250.
- *Fix:* Use smoothing, binning, or remove outliers.

4. Inconsistency Contradictory or non-matching data in different places.

- *Example:* The same customer's city written as "Mumbai" in one row and "Bombay" in another; or a date in two different formats.
- *Fix:* Standardize formats and correct conflicts.

Conclusion: Validation finds these issues, and cleaning (handling incompleteness, noise, and inconsistency) fixes them — giving good-quality data for analysis.

4. Explain data reduction in detail with example. ★ ★ ★



Data Reduction means **reducing the size (volume) of data** while keeping its important information. Smaller data is faster to process and store, and gives almost the same results.

Main techniques:

1. **Dimensionality Reduction** — Reduce the number of **columns (attributes)** by removing or combining less useful ones. *Example:* PCA combines many columns into a few.
2. **Numerosity Reduction** — Reduce the number of **rows** by replacing data with a smaller model or summary. *Example:* using a sample instead of all rows.

3. **Data Compression** — Encode data so it takes **less space**. *Example:* ZIP files, image compression.
4. **Feature Selection** — Keep only the **most useful columns** and drop the rest. *Example:* For predicting marks, keep "study hours" and drop "roll number".
5. **Sampling** — Pick a **smaller representative group** from the full data. *Example:* surveying 1,000 people instead of 1 lakh.

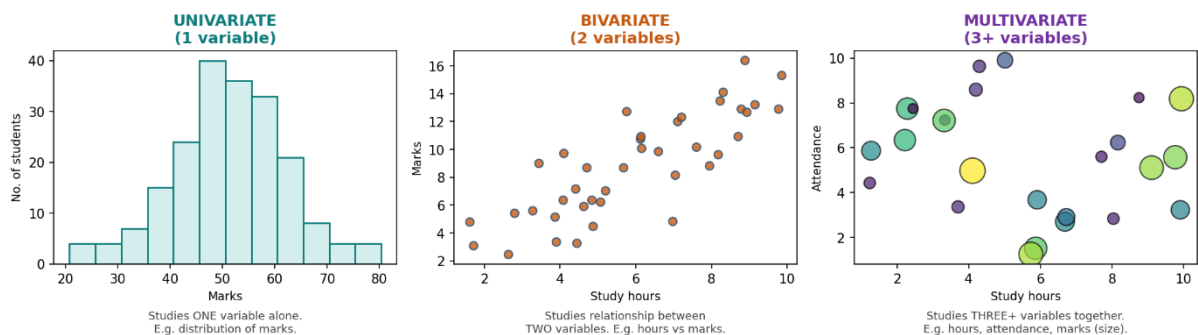
Why useful: Saves time and storage, reduces cost, and makes analysis faster — without losing accuracy.

5. Difference between Univariate, Bivariate, and Multivariate analysis. ★ ★ ★

These three describe analysis based on **how many variables (columns)** we study at a time.

Type	Variables	Meaning	Example
Univariate	1	Study one variable alone	Distribution of student marks
Bivariate	2	Study the relationship between two variables	Study hours vs marks
Multivariate	3 or more	Study many variables together	Hours + attendance + marks

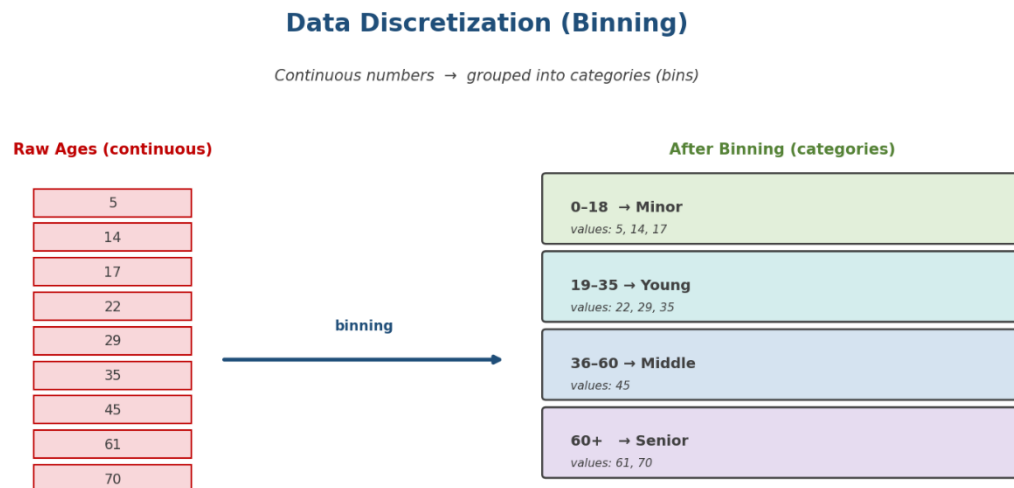
Univariate vs Bivariate vs Multivariate Analysis



- 1. Univariate Analysis** — Looks at a single variable to understand its pattern (average, range, distribution). Uses tools like histograms and bar charts.
- 2. Bivariate Analysis** — Looks at two variables to see if they are related (do they increase together?). Uses scatter plots and correlation.
- 3. Multivariate Analysis** — Looks at three or more variables together to find complex patterns. Uses tools like multiple regression and bubble charts.

Conclusion: As we move from univariate to multivariate, the analysis becomes more detailed and complex but gives a fuller picture.

6. Write a short note on data discretization. ★ ★ ★



Data Discretization means **converting continuous numeric data into a small number of categories (intervals or bins)**. Instead of exact numbers, we use groups.

Example: Ages turned into groups:

- 0–18 → Minor
- 19–35 → Young Adult
- 36–60 → Middle-aged
- 60+ → Senior

So a person aged 27 is simply labeled "Young Adult".

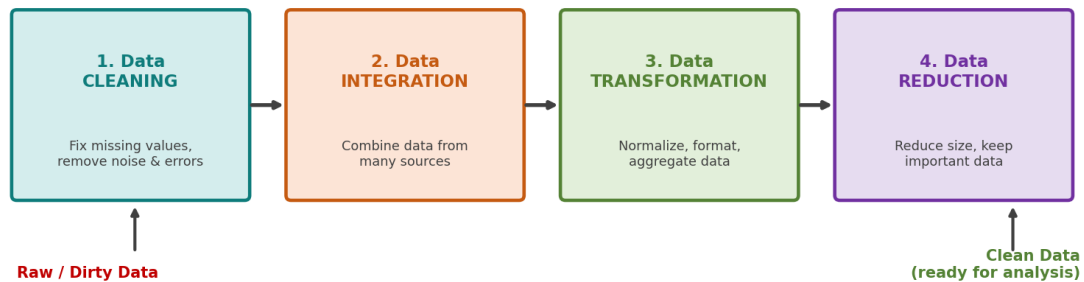
Two main methods:

1. **Equal-width** — divide the range into bins of **equal size** (e.g., 0–20, 20–40, 40–60).
2. **Equal-frequency** — make bins that each have the **same number of records**.

Why useful: It makes data simpler, easier to chart, reduces noise, and helps in grouping and summarizing for reports.

7. Discuss the need for data pre-processing and any 2 techniques used.

Data Pre-processing Steps



Need for data pre-processing: Real-world data is usually **dirty** — it has missing values, errors, and inconsistencies. If we analyze such data directly, we get **wrong results**. Pre-processing cleans and prepares the data so analysis is accurate and reliable. (Rule: "Garbage in, garbage out.")

Two important techniques:

1. **Data Cleaning** — Fixes data-quality problems: fills missing values, removes noise and errors, and corrects inconsistencies. *Example:* filling a blank "Age" with the average age.
2. **Data Transformation** — Converts data into a suitable form: normalization (scaling), aggregation (summarizing), and discretization (grouping). *Example:* scaling salary and age to the same 0–1 range.

(Other steps are Data Integration — combining sources, and Data Reduction — reducing size.)

Conclusion: Pre-processing improves data quality, which directly improves the quality of decisions made from it.

8. Explain Dimensionality Reduction and Data Compression. ★

Both are **data reduction** techniques used to make data smaller and easier to handle.

1. Dimensionality Reduction Means **reducing the number of columns (attributes/features)** in the data, by removing unimportant ones or combining them.

- *Example:* A dataset with 100 columns is reduced to 10 important ones using **PCA**.
- *Benefit:* Removes redundant data, speeds up analysis, and avoids confusion from too many features.

2. Data Compression Means **encoding data so it takes up less storage space**, then decoding it when needed.

- *Types:* **Lossless** (no data lost, e.g., ZIP files) and **Lossy** (some detail lost, e.g., JPEG images).
- *Benefit:* Saves storage and makes data transfer faster.

Difference: Dimensionality reduction reduces the **number of features**; data compression reduces the **storage size** of the data.

9. Define dirty data. What are the reasons for dirty data?

Dirty Data means data that is **incorrect, incomplete, or inconsistent** — data that cannot be trusted for analysis as it is.

Common reasons for dirty data:

1. **Human errors** — wrong typing or spelling mistakes during entry.
2. **Missing values** — fields left blank.
3. **Duplicate records** — the same data entered more than once.
4. **Inconsistent formats** — dates or names written differently (e.g., "Mumbai" vs "Bombay").
5. **Faulty devices/sensors** — wrong readings recorded automatically.
6. **Data integration issues** — errors when combining data from different sources.
7. **Outdated data** — old information that is no longer correct.

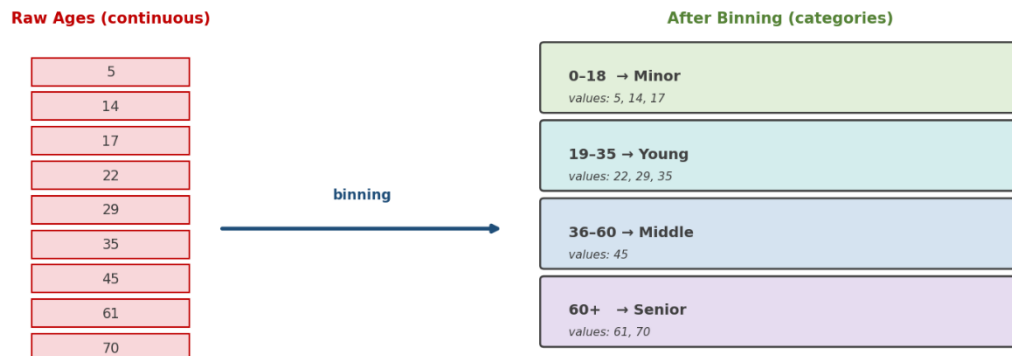
Example: A customer table where one age is "-5", one phone number is missing, and the same person appears twice.

Conclusion: Dirty data must be cleaned before analysis, otherwise it gives wrong and misleading results.

10. Explain the working of binning with a suitable example

Data Discretization (Binning)

Continuous numbers → grouped into categories (bins)



Binning means **grouping continuous values into a few intervals called "bins"**, and then often replacing values to remove noise (smoothing).

Steps:

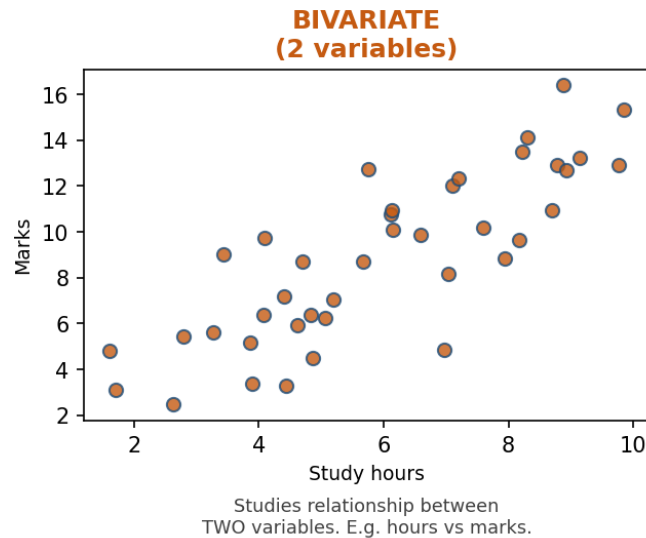
1. **Sort** the data.
2. **Divide** it into bins (equal-width or equal-frequency).
3. **Smooth** each bin using one of these methods:
 - **By mean** — replace all values in a bin with the bin's average.
 - **By median** — replace with the middle value.
 - **By boundary** — replace each value with the nearest bin edge.

Example: Data = 4, 8, 15, 21, 21, 24, 25, 28, 34

- Bin 1: 4, 8, 15 → smoothing by mean (9) → 9, 9, 9
- Bin 2: 21, 21, 24 → mean (22) → 22, 22, 22
- Bin 3: 25, 28, 34 → mean (29) → 29, 29, 29

Why useful: Reduces noise, simplifies data, and turns numbers into easy categories for analysis and reports.

11. What is Bivariate analysis? Why is it important? Discuss the different types with example



Bivariate Analysis means **studying two variables together** to find out **how they are related** — whether they increase together, decrease together, or have no link.

Why it is important:

- Helps find **relationships** between two factors.
- Useful for **prediction** (e.g., if study hours rise, do marks rise?).
- Helps in **decision-making** by showing cause-and-effect links.

Types of bivariate analysis:

1. **Numeric vs Numeric** — both are numbers. *Example:* Study hours vs Marks → shown using a **scatter plot** and correlation.
2. **Numeric vs Categorical** — one number, one category. *Example:* Salary by Gender → shown using **bar charts** or group averages.
3. **Categorical vs Categorical** — both are categories. *Example:* Gender vs "Likes Coffee" → shown using a **contingency table**.

Example: Plotting advertising spend (X) against sales (Y); if the dots rise upward, more advertising leads to more sales (positive relationship).

12. What is a Contingency Table? What is Marginal Distribution? Justify with a suitable example. ★ ★

Contingency Table & Marginal Distribution

Example: Gender vs Likes Coffee (Yes / No)

	Coffee = Yes	Coffee = No	TOTAL (row)
Male	30	10	40
Female	20	40	60
TOTAL (col)	50	50	100

- Inside cells = joint counts (e.g. 30 males like coffee)
- Orange row/column TOTALS = Marginal Distribution

Contingency Table A table that shows the relationship between two categorical variables by counting how often each combination occurs. It is also called a **cross-tabulation**.

Example: Gender (Male/Female) vs Likes Coffee (Yes/No):

	Coffee = Yes	Coffee = No	Total
Male	30	10	40
Female	20	40	60
Total	50	50	100

The inside cells show **joint counts** (e.g., 30 males like coffee).

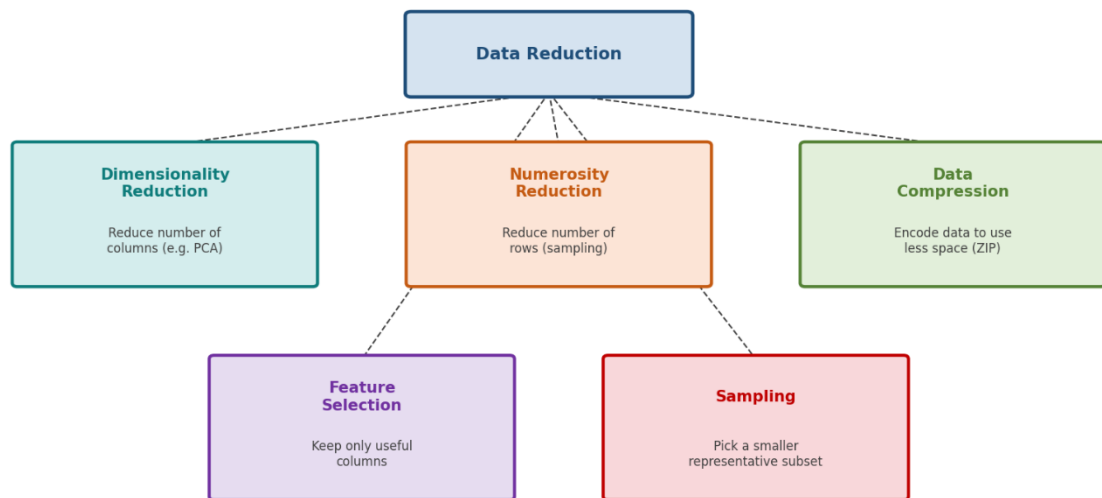
Marginal Distribution The **totals at the edges (margins)** of the table — the row totals and column totals. They show the distribution of **one variable alone**, ignoring the other.

- Row margins: Male = 40, Female = 60 (gender distribution).
- Column margins: Yes = 50, No = 50 (coffee distribution).

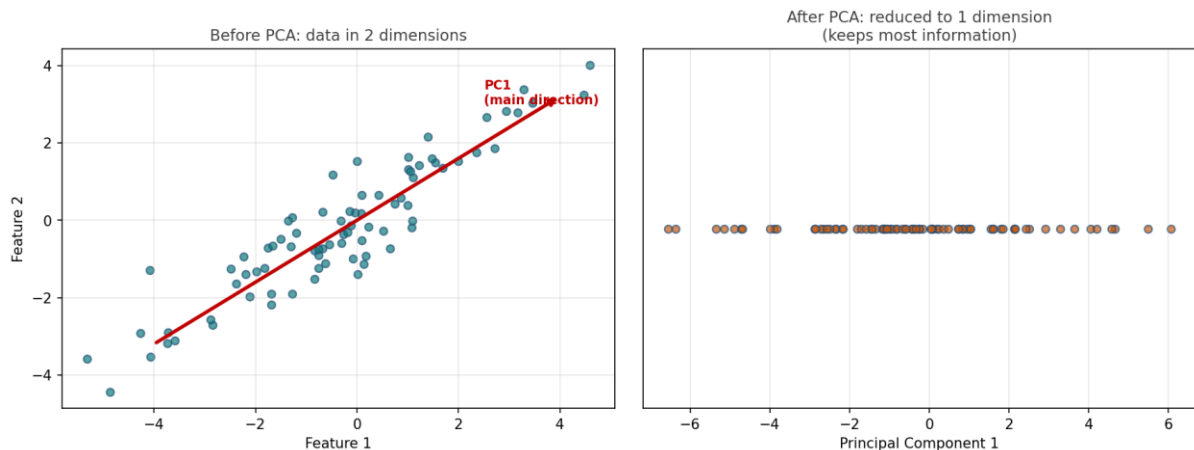
Conclusion: The contingency table shows the relationship between two variables, and the marginal distribution shows the total for each variable separately.

13. Explain Data Reduction technique: Sampling, Feature Selection, Principal Component Analysis (PCA). ★ ★

Data Reduction Techniques



Principal Component Analysis (PCA)



These are three **data reduction** techniques.

1. Sampling — Selecting a **smaller representative group** from the full dataset, instead of using all the data.

- *Example:* Surveying 1,000 customers to understand 1 lakh customers.
- *Benefit:* Saves time and cost while giving close results.

2. Feature Selection — Choosing only the **most useful columns (features)** and removing the unimportant ones.

- *Example:* To predict marks, keep "study hours" and "attendance"; drop "roll number".
- *Benefit:* Simpler, faster, and more accurate models.

3. Principal Component Analysis (PCA) — A technique that **combines many related columns into a few new columns** (called principal components) that still hold most of the information.

- *Example:* Reducing 50 features to 5 main components.
- *Benefit:* Removes redundancy and reduces dimensions while keeping key information.

Difference: Sampling reduces **rows**, while feature selection and PCA reduce **columns**.

14. How Mean, Median and Mode are Used During Data Cleaning

When data has **missing values**, these three measures (called measures of central tendency) are used to **fill the gaps** with a sensible value.

1. **Mean (average)** — Used to fill missing **numeric** values when data is evenly spread (no extreme values).
 - *Example:* A blank "marks" cell filled with the class average.
2. **Median (middle value)** — Used to fill missing numeric values when data has **outliers**, because the median is not affected by extreme values.
 - *Example:* Missing salary filled with the median salary (better than mean if a few salaries are very high).
3. **Mode (most frequent value)** — Used to fill missing **categorical** values.
 - *Example:* A blank "city" cell filled with the most common city.

Why useful: These methods fill missing data quickly and reasonably, so the dataset stays complete and usable without deleting records.

15. Explain the need of data cleaning and different methods

Need for data cleaning: Real data has errors, missing values, and inconsistencies. Cleaning is needed so that the data is **accurate, complete, and reliable** — leading to correct analysis and good decisions.

Methods of data cleaning:

1. **Handling Missing Values** — fill with mean/median/mode, fill manually, or remove the record.
2. **Removing Noise** — use **binning, smoothing**, or remove outliers.
3. **Removing Duplicates** — delete repeated records.

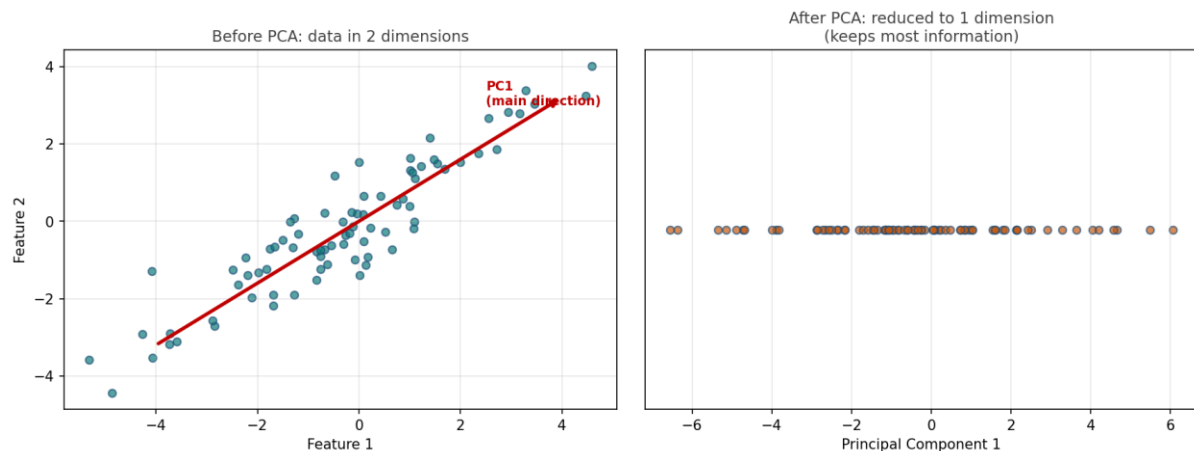
4. **Fixing Inconsistencies** — standardize formats (e.g., one date format, one spelling of a city).
5. **Correcting Errors** — fix wrong entries (e.g., age = -5).

Example: In a customer table, we fill blank phone numbers, correct "Bombay" to "Mumbai", and delete a duplicate row.

Conclusion: Clean data is the foundation of good BI — without it, even the best analysis gives wrong results.

16. Explain the working of PCA and its applications

Principal Component Analysis (PCA)



PCA (Principal Component Analysis) is a technique that **reduces the number of columns (dimensions)** by combining related columns into a few new columns called **principal components**, which still hold most of the information.

Simple working steps:

1. **Standardize** the data so all features are on the same scale.
2. Find how features are **related** to each other (correlation).
3. Find the **main directions** where the data varies the most — these become the principal components.
4. **Keep the top few components** (which hold most information) and drop the rest.
5. The data is now represented with **fewer columns**.

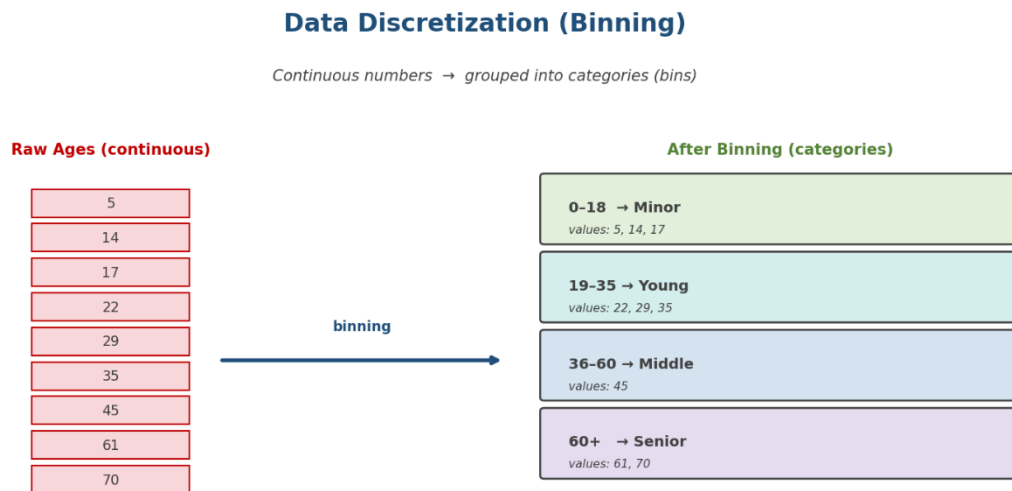
Simple idea: If two columns carry similar information, PCA merges them into one — reducing size while keeping the meaning. (See diagram: 2D data is reduced to 1D along its main direction.)

Applications:

- **Image compression** (reducing image size).

- **Face recognition** systems.
- **Reducing features** before machine learning.
- **Data visualization** (showing high-dimensional data in 2D).
- **Removing noise** and redundant data.

17. Explain data discretization. What are the two methods?



Data Discretization means **converting continuous numeric data into categories (intervals/bins)**, so exact numbers are replaced by groups.

Two main methods:

1. **Equal-Width Binning** — The full range is divided into bins of **equal size**.
 - *Example:* Ages 0–60 split into 0–20, 20–40, 40–60 (each width = 20).
 - Simple, but some bins may have very few or very many records.
2. **Equal-Frequency Binning** — Bins are made so that **each bin has the same number of records**.
 - *Example:* If there are 90 records, make 3 bins of 30 records each.
 - Balanced, but bin widths may be unequal.

(Discretization can also be **supervised**, using class labels, or **unsupervised**, without them.)

Why useful: Makes data simpler, reduces noise, and converts numbers into easy categories for analysis and reporting.

18. Write short notes on Data Validation.

Data Validation is the process of **checking that data is correct, complete, and in the proper format** before it is stored or analyzed. It ensures the data is trustworthy.

Key points:

- It is done **at the time of data entry or import**.
- **Common checks:**
 - **Type check** — a number field has only numbers.
 - **Range check** — value is within limits (e.g., age 0–120).
 - **Format check** — correct pattern (e.g., email has "@").
 - **Mandatory check** — required fields are not blank.
 - **Consistency check** — data does not contradict other data.
- *Example:* Rejecting an entry where "Age = 200" or an email without "@".

Why useful: Validation stops dirty data from entering the system, which keeps analysis accurate and decisions reliable.